

Syllabus_B.Sc. (Hons.) Agriculture, Pt-I, Semester-II

HORT-121	Post Harvest Processing Technology*	2(0+2)
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Objectives

1. To provide knowledge on postharvest processing technology of various horticultural produce.
1. To provide knowledge on value addition of horticultural crops.
2. To provide knowledge on post-harvest management of horticultural crops.

Practical

Practice in judging the maturity of various horticultural produce, Determination of physiological loss in weight and quality. Grading of horticultural produce, post-harvest treatment of horticultural crops. Packaging studies in fruits, vegetables, plantation crops and cut flowers by using different packaging materials and Storage methods. Post-harvest disorders in horticultural produce. Equipment used in post-harvest technology processing units. Canning of fruits and vegetables. Preparation of squash, RTS, cordial, syrup, jam, jelly and marmalade. Preparation of candies, preserves, chutneys, sauces and pickles (hot and sweet). Dehydration of fruits and vegetables. Visit to markets/processing unit/packaging houses/cold storage units.

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PPATH-121 PRODUCTION TECHNOLOGY OF BIOAGENTS

2 (0+2)

Objectives

1. To study principle and application of eco friendly and sustainable management strategies of plant diseases.

Practical:

Concept of biological control, definitions, importance, principles of plant disease management with bioagents, history of biological control, merit and demerits of biological control. Types of biological interactions, competition: mycoparasitism, exploitation for hypovirulence, rhizosphere colonization, competitive saprophytic ability, antibiosis, induced resistance, mycorrhizal associations, operational mechanism and its relevance in biological control. Factors governing biological control, role of physical environment, agro ecosystem, operational mechanisms and cultural practices in biological control of pathogens, pathogens and antagonists and their relationship, bio control agents, comparative approaches to biological control of plant pathogens by resident and introduced antagonists, control of soil-borne and foliar diseases. Compatibility of bioagents with agrochemicals and other antagonistic microbes. Commercial production of antagonists, their delivery systems, application and monitoring, biological control in IDM, IPM and organic farming system, biopesticides available in market. Quality control system of biocontrol agents. Isolation, characterization and maintenance of antagonists, methods of study of antagonism and antibiosis, application of antagonists against pathogen *in vitro* and *in vivo* conditions. Preparation of different formulations of selected bioagents and their mass production. Quality parameters of biocontrol agents. One week exposure visit to commercial bio control agents production unit.

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PPATH-122 FUNDAMENTALS OF PLANT PATHOLOGY

3(2+1)

Objectives

1. To get acquainted with the role of different microorganisms in the development of plant disease
2. To get general concepts and classification of plant diseases
3. To get knowledge of general characteristics of fungi, bacteria, virus, and other microorganisms causing plant diseases
4. To acquaint the students with reproduction in fungi, and bacteria, causing plant diseases
5. To get acquainted with various plant disease management principles and practices

Theory:

Introduction to Plant Pathology: Concept of disease in plants, Importance of Plant disease, Scope & Objective of Plant pathology, History of Plant Pathology with special references to India; Terms & concepts in Plant Pathology Pathogenesis. Causes and classification of plant diseases, diseases and symptoms due to biotic/ abiotic causes. Parasitism and pathogenesis; Development of disease in plants: Disease Triangle, Disease cycle. Fungi: general Characters, Somatic structures, Types of fungi thalli, fungi tissues, modification of thallus and their morphology, reproduction (Sexual & Asexual) Nomenclature, Binomial system of nomenclature, Classification of fungi (key to Domain to Phylum). Bacteria and mollicutes: general morphological characters. Reproduction and classification of plant pathogenic bacteria. Important plant pathogens: Mollicutes; Flagellant protozoa; FVB; Green algae and parasitic higher plants, prions & viroids Viruses : Nature , Structure and Transmission Role of enzymes and toxins in disease development. Defense mechanism in plants. Principles of Plant disease management. Disease management with chemicals, Host resistance, cultural and biological method. Principals of IDM & modules for What, Rice, Sugarcane, Cotton, groundnut, citrus and chickpea.

Practical

Study of the microscope; Acquaintance with laboratory material and equipment; Study of different plant disease symptoms; Microscopic examination of general structure of fungi; Simple staining of bacteria: Direct and indirect staining, Gram staining of bacteria; Microscopic examination of fungal disease specimen; Microscopic examination of bacterial disease specimen; Preparation of culture media; Isolation of plant pathogens: Fungi, bacteria and viruses; Purification of plant pathogens; Study on plant disease diagnosis: Koch's Postulates, Characteristics, formulation, methods of application and calculation on fungicides.

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AGEXT- 121

PERSONALITY DEVELOPMENT

2 (1+1)

Objective

To make students realize their potential strengths, cultivate their inter-personal skills and improve employability.

Theory

Personality Definition, Nature of personality, theories of personality and its types. The humanistic approach - Maslow's self-actualization theory, shaping of personality, determinants of personality, Myers-Briggs Typology Indicator, Locus of control and performance, Type A and Type B Behaviours, personality and Organizational Behaviour. Foundations of individual behavior and factors influencing individual behavior, Models of individual behavior, Perception and attributes and factors affecting perception, Attribution theory and case studies on Perception and Attribution. Learning: Meaning and definition, theories and principles of learning, Learning and organizational behavior, Learning and training, learning feedback. Attitude and values, Intelligence- types of Intelligence, theories of intelligence, measurements of intelligence, factors influencing intelligence, intelligence and Organizational behavior, emotional intelligence. Motivation- theories and principles, Teamwork and group dynamics.

Practical

MBTI personality analysis, Learning Styles and Strategies, Motivational needs, Firo-B, Inter personal Communication, Teamwork and team building, Group Dynamics, Win-win game, Conflict Management, Leadership styles, Case studies on Personality and Organizational Behavior.

Objectives

1. To know the history of entomology, classification of insects and their relationship with other arthropods
2. To study the various morphological characters of class insect and their importance for classification of insects
3. To get an idea about the different physiological systems of insects and their roles in growth and development and communications of insects
4. To study the characteristics of commonly observed insect orders and their economically important families

Theory

History of Entomology in India. Major points related to dominance of Insects in Animal kingdom. Classification of phylum Arthropoda up to classes. Relationship of class Insects with other classes of Arthropoda. Morphology: Structure and functions of insect cuticle and molting. Body segmentation. Structure of head, thorax and abdomen. Structure and modifications of insect antennae, mouth parts, legs, Wing venation, modifications and wing Structure and functions of digestive, circulatory, excretory, respiratory, nervous, secretory (Endocrine) and reproductive system, in insects. Types of reproduction in insects. Systematics: Taxonomy - importance, history and development and binomial nomenclature. Definitions of Biotype, Sub- species, Species, Genus, Family and Order. Classification of class Insecta up to Orders, basic groups of present day insects with special emphasis to orders and families of Agricultural importance like Orthoptera: Acrididae, Tettigoniidae, Gryllotalpidae; Chrysopidae; Lepidoptera: Papilionidae, Noctuidae, Sphingidae, Pyralidae, Gelechiidae, Arctiidae, Saturniidae, Bombycidae; Coleoptera: Coccinellidae, Chrysomelidae, Cerambycidae, Curculionidae, Bruchidae, Scarabaeidae; Hymenoptera: Tenthredinidae, Apidae. Trichogrammatidae, Ichneumonidae, Braconidae, Chalcididae; Diptera: Cecidomyiidae, Tachinidae, Agromyziidae, Culicidae, Muscidae, Tephritidae.

Practical

Methods of collection and preservation of insects including immature stages; External features of Grasshopper/Blister beetle; Types of insect antennae, mouthparts and legs; Wing venation, types of wings and wing coupling apparatus. Types of insect larvae and pupae;

Dissection of digestive system in insects (Grasshopper); Study of characters of orders Orthoptera, Dictyoptera, Odonata, Isoptera, Thysanoptera, Hemiptera, Lepidoptera, Neuroptera, Coleoptera, Hymenoptera, Diptera and their families of agricultural importance.

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ESDM-121 ENVIRONMENTAL STUDIES AND DISASTER MANAGEMENT 3 (2+1)

Objective

- To expose and acquire knowledge on the environment and to gain the state-of-the-art skill and expertise on management of disasters

Theory

Introduction to Environment-

Environmental studies: Definition, scope and importance- Multidisciplinary nature of environment studies-Segments of Environment-Spheres of Earth -Lithosphere- Hydrosphere-Atmosphere-Different layers of atmosphere. Natural Resources:

Classification - Forest resources. Water resources. Mineral resources Food resources. Energy resources. Land resources. Soil resources. Ecosystems: Concept of an ecosystem - Structure and function of an ecosystem - Energy flow in the ecosystem. Types of ecosystem. Biodiversity and its conservation: Introduction, definition, types. Biogeographical classification of India. Importance and Value of biodiversity. Biodiversity hot spots. Threats and Conservation of biodiversity.

Environmental Pollution: Definition, cause, effects and control measures of: a. Air pollution.

b. Water pollution. c. Soil pollution. d. Marine pollution. e. Noise pollution. f. Thermal pollution g. Light pollution. Solid Waste Management: Classification of solid wastes and management methods, Composting, Incineration, Pyrolysis, Biogas production, Causes, effects and control measures of urban and industrial wastes. Social Issues and the Environment: Urban problems related to energy. Water conservation, rain water harvesting, watershed management. Environmental ethics: Issues and possible solutions, climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Environment Protection Act. Air (Prevention and Control of Pollution) Act. Water (Prevention and control of Pollution) Act. Wildlife Protection Act. Forest Conservation Act. Human Population and the Environment: Environment and human health: Human Rights, Value Education. Women and Child Welfare. Role of Information Technology in Environment and human health.

Disaster management: Disaster definition - Types - Natural Disasters - Floods, drought, cyclone, earthquakes, landslides, avalanches, volcanic eruptions, Heat and cold waves. Man Made Disasters: Nuclear disasters, chemical disasters, biological disasters, building fire, coal fire, forest fire, oil fire, road accidents, rail accidents, air accidents, sea accidents. International and National strategy for disaster reduction. Concept of disaster management, national disaster management framework; financial arrangements; role of NGOs, community-based organizations and media in disaster management. Central, state, district and local administration in disaster control; Armed forces in disaster response; Police and other organizations in disaster management.

Practical

Visit to a local area to document environmental assets river/forest/grassland/hill/mountain. Energy: Biogas production from organic wastes. Visit to wind mill / hydro power / solar power generation units. Biodiversity assessment in farming system. Floral and faunal diversity assessment in polluted and un polluted system. Visit to local polluted site - Urban/Rural/Industrial/Agricultural to study of common plants, insect and birds. Environmental sampling and preservation. Water quality analysis: pH, EC and TDS. Estimation of Acidity, Alkalinity. Estimation of water hardness. Estimation of DO and BOD in water samples. Estimation of COD in water samples. Enumeration of *E. coli* in water sample. Assessment of Suspended Particulate Matter (SPM). Study of simple ecosystem –Visit to pond/river/hills. Visit to areas affected by natural disaster.

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LPM-121

LIVESTOCK AND POULTRY MANAGEMENT

2(1+1)

Objectives:

1. Provide basic knowledge to the students about scientific livestock and poultry rearing practices.
2. Entrepreneurship development through livestock/poultry and agriculture integrated farming system.

Theory

Role of livestock in the national economy. Reproduction in farm animals and poultry. Housing principles, space requirements for different species of livestock and poultry. Management of calves, growing heifers and milch animals. Management of sheep, goat, camel and swine. Incubation, hatching and brooding. Management of growers and layers. Important Indian and exotic breeds of cattle, buffalo, sheep, goat, swine, camel and poultry. Improvement of farm animals and poultry. Digestive system of livestock and poultry. Classification of feed and fodder. Nutrients and their functions. Feed ingredients for ration for live stock and poultry. Feed supplements and feed additives. Feeding of livestock and poultry. Different diseases of livestock and poultry, their prevention and control.

Practical

External body parts of cattle, buffalo, sheep, goat, swine, camel and poultry. Handling and restraining of livestock & poultry. Identification methods of farm animals and poultry. Dentition of livestock. Visit to livestock and poultry farms. Daily routine farm operations and farm records of livestock and poultry farms. Judging of cattle, buffalo and poultry. Culling of livestock and poultry. Planning and layout of housing for different types of livestock & poultry. Computation and formulation of ration for livestock and poultry. Proximate principles of feed. Clean milk production, milking methods. Hatchery management and poultry farm equipment's. Management of chicks, growers and layers. Debeaking, dusting and vaccination. Economics of cattle, buffalo, sheep, goat, swine and poultry production.

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SSAC-121

SOIL FERTILITY MANAGEMENT

3 (2+1)

Objective

To provide a comprehensive knowledge of soil fertility, plant nutrition, fertilizers, and nutrient management

Theory

History of soil fertility and plant nutrition. criteria of essentiality. Role, deficiency and toxicity symptoms of essential plant nutrients, Mechanisms of nutrient transport to plants, factors affecting nutrient availability to plants. Chemistry of macro and micronutrients. Soil fertility evaluation, Soil testing. Critical levels of different nutrients in soil. Forms of nutrients in soil, plant analysis, rapid plant tissue tests. Indicator plants. Introduction and importance of manures and fertilizers. Fertilizer recommendation approaches.

Integrated nutrient management. Chemical fertilizers: classification, composition and properties of major fertilizers, secondary and micronutrient fertilizers, Complex fertilizers, Customized fertilizers, water soluble fertilizers nano fertilizers Soil amendments, Fertilizer Storage, Fertilizer Control Order. Methods of fertilizer recommendations to crops. Factor influencing nutrient use efficiency (NUE), methods of application under rainfed and irrigated conditions. STCR/RTNM/IPNS, Carbon sequestration and Carbon Trading, Preparation and properties of major manures (FYM, Compost, Vermicompost, Green manuring, Oilcakes).

Practical

Introduction of analytical instruments and their principles, calibration and applications of Coloremety and flame photometry; Estimation of soil organic carbon Estimation of alkaline hydrolysable N in soils; Estimation of soil extractable P in soils; Estimation of exchangeable K in soils; Estimation of exchangeable Ca and Mg in soils; Estimation of soil extractable S in soils; Estimation of DTPA extractable Zn in soils; Estimation of N in plants; Estimation of P in plants; Estimation of K in plants; Estimation of S in plants.